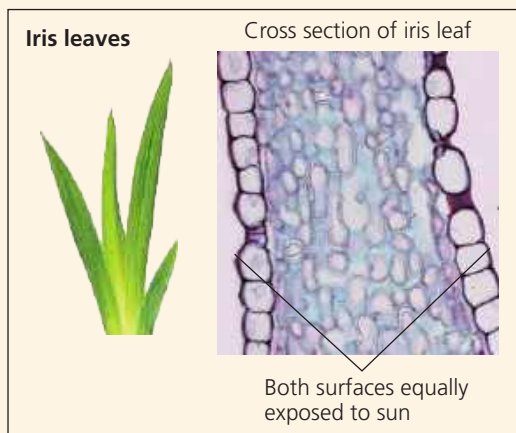
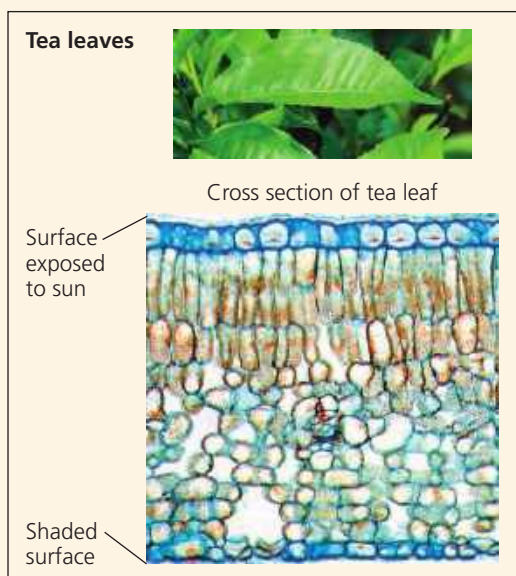


- 12. VISUAL SKILLS** How does the internal anatomy of a tea leaf differ from that of an iris leaf? (Review Figure 35.18.) How might this difference relate to the orientations of the leaves? Explain how the difference shows that structure fits function.

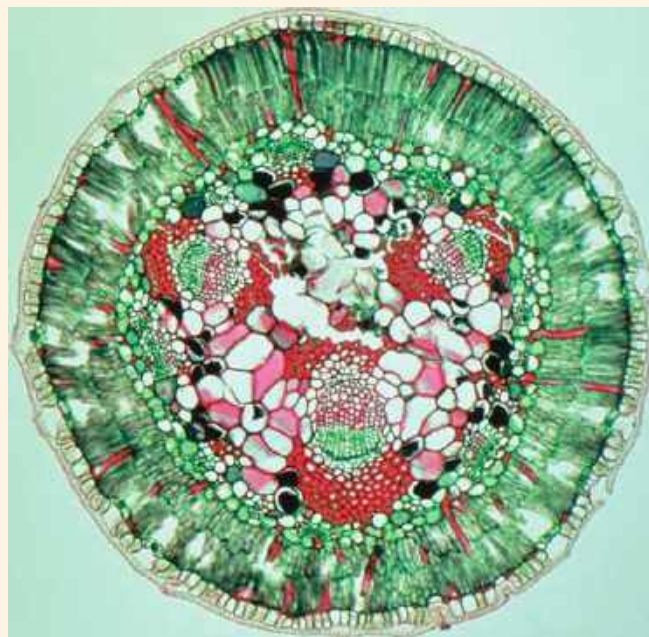


- 13. VISUAL SKILLS** Visitors to the state of Washington can marvel at an unusual sight: a bicycle embedded 4 m above the ground in a living tree. Use your knowledge of primary and secondary growth to visualize and explain how this happened.



## Levels 5-6: Evaluating/Creating

- 14. EVOLUTION CONNECTION** Evolutionary biologists have coined the term *exaptation* to describe a common occurrence in the evolution of life: A limb or organ evolves in a particular context but over time takes on a new function (see Concept 25.6). What are some examples of exaptations in plant organs?
- 15. SCIENTIFIC INQUIRY** Grasslands typically do not flourish when large herbivores are removed. Instead, grasslands are replaced by broad-leaved herbaceous eudicots, shrubs, and trees. Based on your knowledge of the structure and growth habits of monocots versus eudicots, suggest a reason why.
- 16. SCIENCE, TECHNOLOGY, AND SOCIETY** Hunger and malnutrition are urgent problems for many poor countries, yet plant biologists in wealthy nations have focused most of their research efforts on *Arabidopsis thaliana*. Some people have argued that if plant biologists are truly concerned about fighting world hunger, they should study cassava and plantain because these two crops are staples for many of the world's poor. If you were an *Arabidopsis* researcher, how might you respond to this argument?
- 17. WRITE ABOUT A THEME: ORGANIZATION** In a short essay (100–150 words), explain how the evolution of lignin affected vascular plant structure and function.
- 18. SYNTHESIZE YOUR KNOWLEDGE**



This stained light micrograph shows a cross section through a plant organ from *Hakea purpurea*, a shrub native to some arid regions of Australia. (a) Review Figures 35.14, 35.17, and 35.18 to identify whether this is a root, stem, or leaf. Explain your reasoning. (b) How might this organ be an adaptation for dry conditions?

For selected answers, see Appendix A.

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